Thermoelectric-Assisted Regeneration of Aminosilica Sorbents for Cookstove Aldehyde Capture

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A computational study testing whether cookstove waste heat can support thermal regeneration of aldehyde-capturing sorbents.

Motivation and Research Question

Biomass cookstove exhaust contains carbonyl VOCs such as formaldehyde and acetaldehyde. Aminosilica sorbents can bind these compounds, but any practical filter has to recover its capacity over repeated cooking cycles.

Research question

Can waste heat harvested through a thermoelectric pathway provide enough local heating to regenerate the sorbent without external power?

1. Polluted exhaust

Cookstove emissions carry VOCs, particulates, CO, and carbonyl species.

2. Saturated sorbent

Amine sites bind aldehydes but capacity is finite.

3. Regeneration barrier

Restoring capacity requires heat, which is difficult to add in low-resource cooking settings.

System Architecture

The proposed filter couples three functions: pollutant capture in an aminosilica bed, thermoelectric energy harvesting from the exhaust temperature gradient, and resistive heating for regeneration.

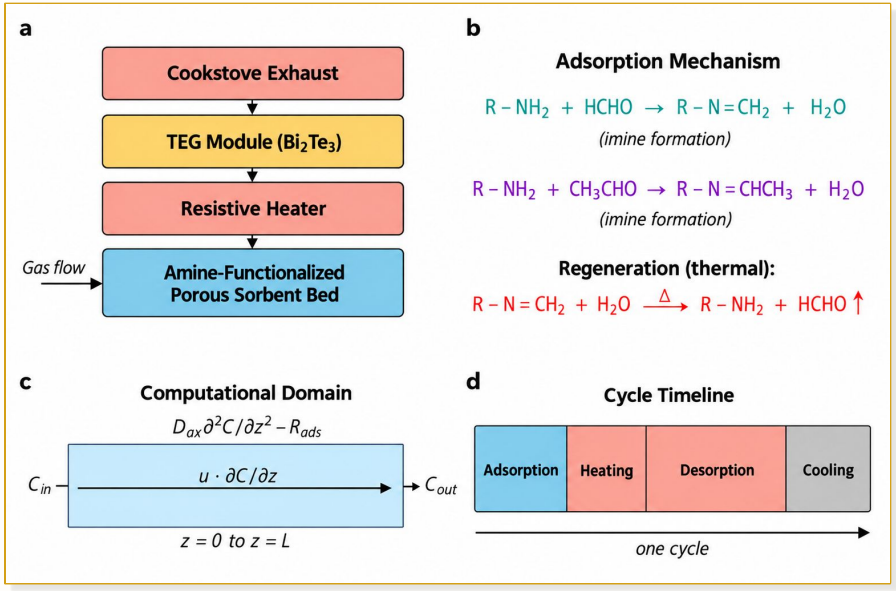


Figure 1. System concept, adsorption mechanism, computational domain, and cycle timeline

Gas path
Aldehyde-containing exhaust passes through a porous sorbent bed.

Thermal path
Hot exhaust and ambient cooling create a temperature gradient across the TEG.

Electrical path
The TEG output is routed to local resistive heating during regeneration.

Design conflict
The system has to capture pollutants while keeping pressure drop low and delivering enough heat to reset the bed.

Computational Model

The model links transport, adsorption, thermal regeneration, thermoelectric output, and pressure drop. I used the simulation results to test whether capture and regeneration can be satisfied at the same time.

Transport
1D axial packed-bed species balance

Sorption
Kinetic uptake and thermal desorption

Thermal + TEG
ΔT, voltage, matched-load power

Pressure drop
Packed-bed flow resistance

Model checks
Grid convergence, solver tolerance, TEG power scaling, and pressure-drop scaling.

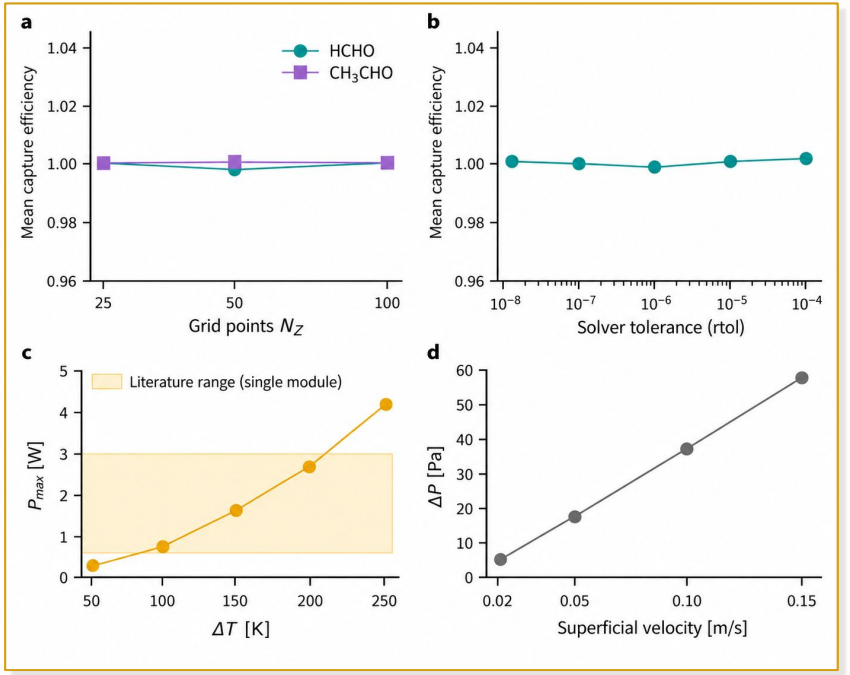


Figure 2. Numerical checks and literature-scale TEG/pressure-drop comparisons

Adsorption Performance

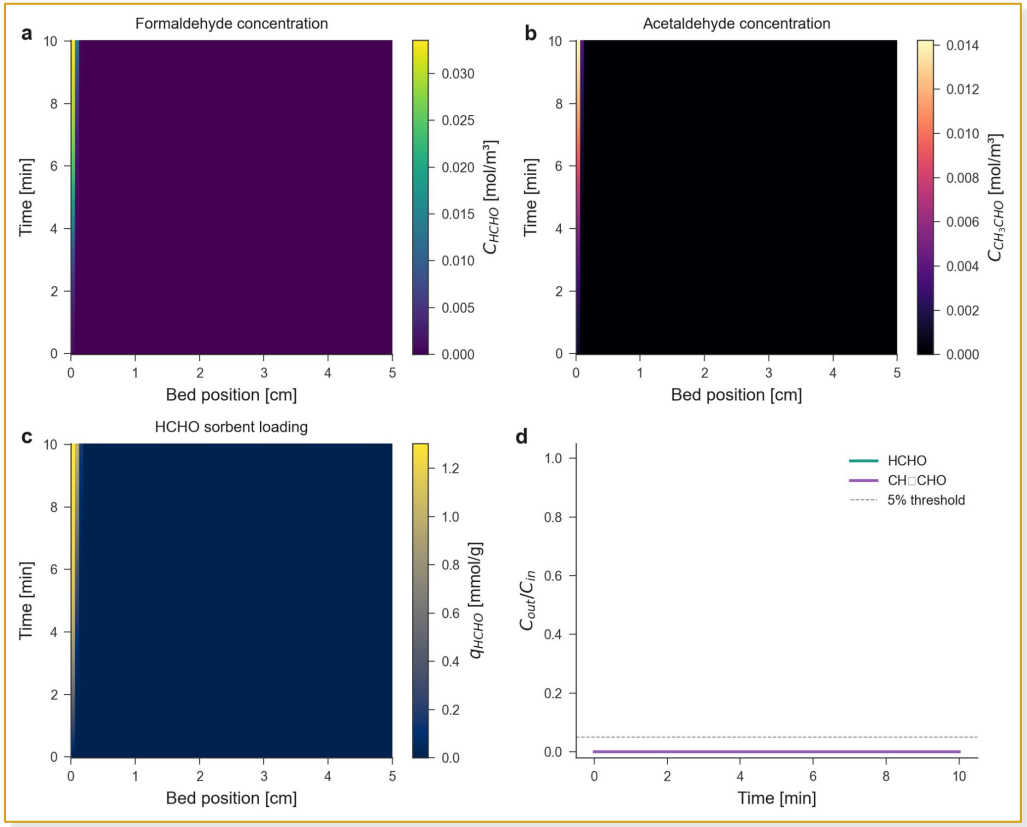


Figure 3. Concentration fields, sorbent loading, and breakthrough behavior

Across the sampled designs, both formaldehyde and acetaldehyde stayed below breakthrough during the simulated adsorption window.

Main result
Modeled capture remained near 100% across the sweep.

Interpretation
The bed was not running out of sorbent capacity during the modeled cooking interval.

Why it matters
Once capture is strong, regeneration energy becomes the deciding constraint.

Regeneration and TEG Performance

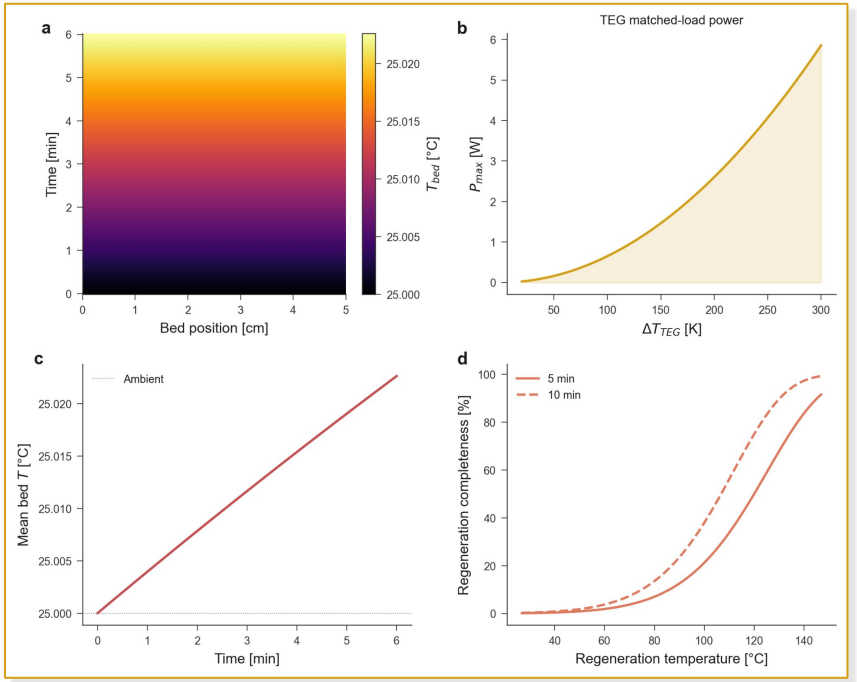


Figure 4. Thermal response and TEG power relationships

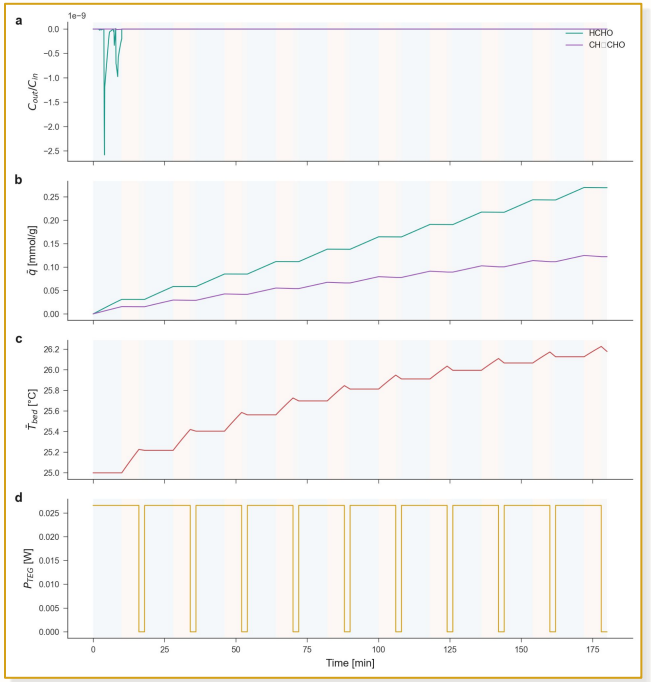


Figure 5. Repeated adsorption/regeneration traces

Single-module TEG output
0.002–0.152 W across 1,500 sweep cases

Energy sufficiency
0 of 1,500 cases fully regenerated the whole bed

Conclusion
Waste heat helps, but this configuration needs localized heating or more usable TEG power.

Design Implications and Future Work

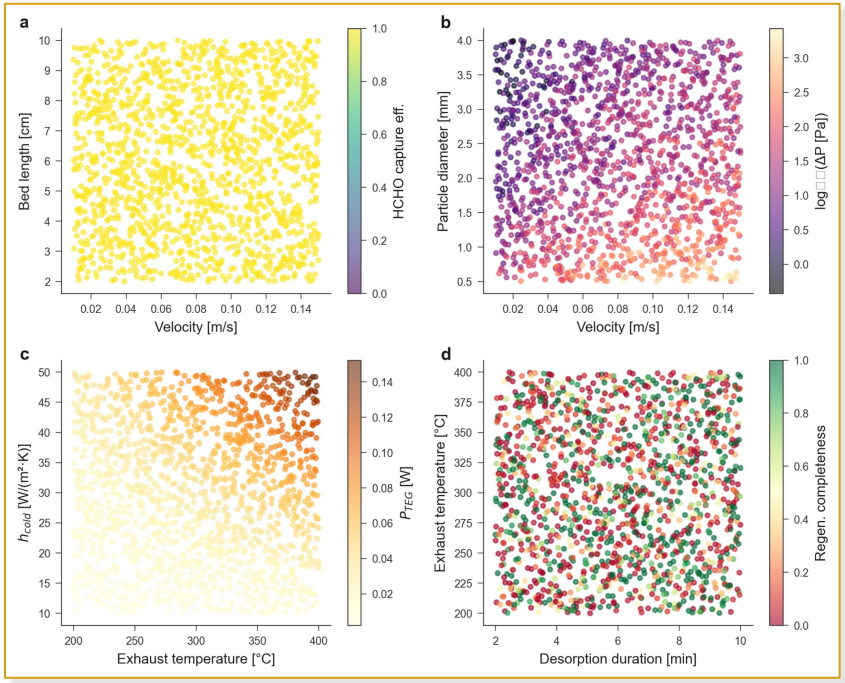


Figure 6. Design-space maps from the parameter sweep

Next steps
Build a small instrumented prototype and use measured temperature, VOC, and pressure-drop data to update the model.

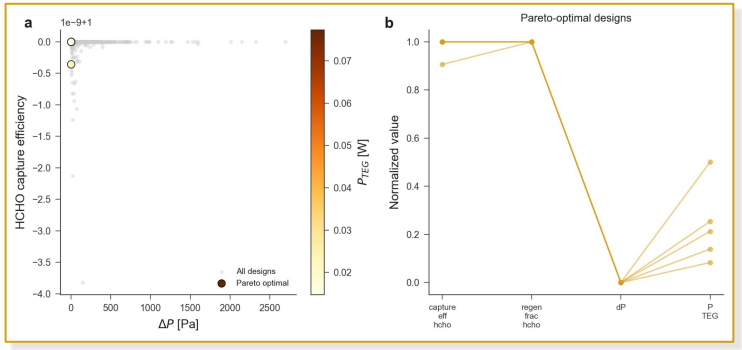


Figure 7. Pareto view of tradeoffs between capture, pressure drop, regeneration, and power

1. Reduce pressure drop
Lower superficial velocity, larger particles, and higher porosity reduce pressure-drop penalties.

2. Localize regeneration
Heat the active sorbent region rather than treating the full bed as one thermal mass.

3. Increase usable ΔT
Improve cold-side heat rejection or use a multi-module TEG path.